

Agentic AI for Healthcare Fraud Detection

Part 1 · Report

Part 2 · Proof of Concept

WHY THIS IS HARD

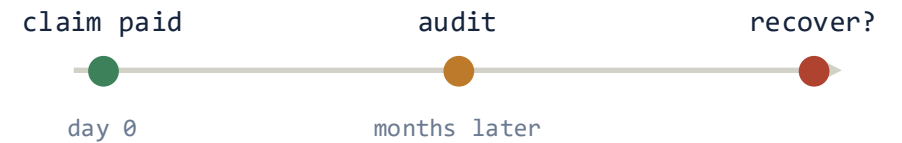
The money leaves **before** anyone checks

SCALE

3–10% of total healthcare spending is lost to fraud, waste, and abuse.

TIMING

Static rule engines and retroactive audits create a **“pay and chase”** cycle: recovering paid dollars is costly and abrades providers.

**Identified ≠ recovered**

A flag is not a dollar back. Denials get appealed; recovery varies by error type.

FROM SINGLE MODELS TO COORDINATED AGENTS

One score is not a decision

SINGLE MODELS & RULE EDITS — WHERE THEY BREAK

- ✗ Recall-tuned classifiers flood investigators with **false positives**; without adversarial checks, errors propagate.

Across 137 studies, supervised tree-based ensemble models and unsupervised models cover unlabeled data but all output “scores”, not decisions tied to reasoning and explanations.

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- ➔ Coordinated roles **mirror the decision process** instead of emitting one number.

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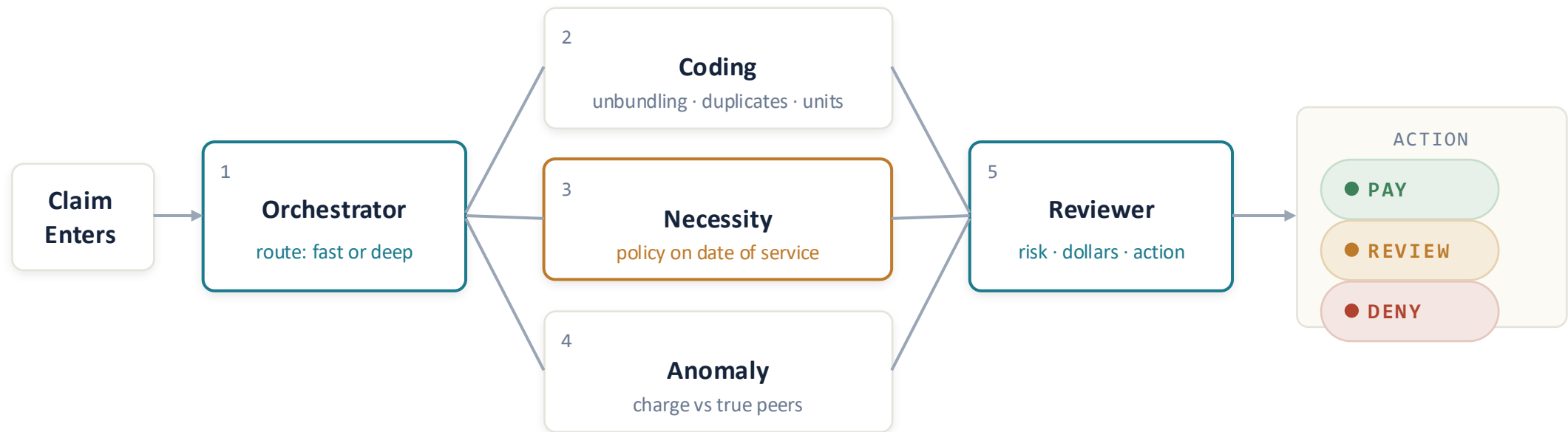
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- Confidence tiers and **traceable reasoning** turn noise into actionable intelligence.

THE PIPELINE

How a claim moves through the proposed agentic system



fast path, simple, low-risk claims skip deep review

The orchestrator triages: routine claims take the fast path; risky ones get the full agent review. The reviewer turns findings into one action and routes large claims to a person.

FIVE AGENTS, FIVE JOBS

Each agent does **one thing** — and shows its work

01

Orchestrator

Estimates a provider's baseline risk and decides how hard to look at the claim.

Bayesian prior + OIG-
LEIE check

02

Coding

Catches coding errors: unbundling, duplicate lines, impossible unit counts.

Deterministic edit
rules (NCCI)

03

Necessity

Checks the service against the coverage policy that applied on the date of service.

Temporal GraphRAG
retrieval

04

Anomaly

Compares the charge to the provider's real peers, not a global average.

$z\text{-score} \cdot \text{specialty} \times \text{code}$

05

Reviewer

Combines findings into a risk score, dollars at risk, and one action.

Weighted scoring +
human review

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THE RETRIEVAL METHOD DECIDES THE DETERMINATION

Why graph retrieval, not plain RAG

LIMIT

Standard RAG returns **unstructured text chunks**, no structured reasoning for a policy call.

GRAPH

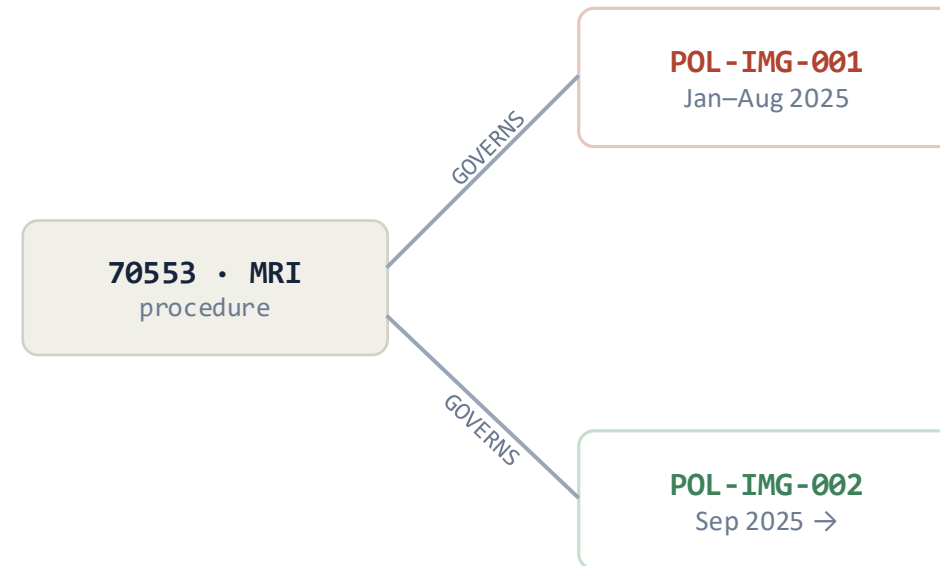
Graph retrieval reasons **across related entities**, procedure, diagnosis, policy and not isolated passages.

MEDICAL

MedGraphRAG returns answers **grounded in evidence with citations**, with higher medical-QA accuracy.

TEMPORAL

Temporal edges anchor retrieval to the **policy in force on the date of service**.



Encoding procedures, diagnoses, and policies as a graph lets a determination cite the applicable rule and its effective period.

BUILT FOR THE PERSON WHO MAKES THE CALL

The system ranks and explains. The reviewer decides.

CLM-100613

DENY

low risk

high risk

\$14,962

BILLED

\$14,962

AT RISK

0.74

RISK

WHY THIS CLAIM IS FLAGGED

POLICY

Service denied under POL-IMG-001, in force on the date of service.

CODING

Lab panel billed as components, not the bundled code.

PEER

Charge is 3.1 times above same-specialty peers for this code.



Queue ranked by dollars at risk, the most consequential claims sit on top.



Plain-language reasons, each tagged to the agent that produced it: policy, coding, or peer.



A disposition exports to a worklist. The human stays in charge of the outcome.

<http://localhost:8501/>

Illustrative console view. Only what was billed, what is recoverable, and why.

STRATEGIC OPTIONS FOR COTIVITI

Three Options

OPTION 1

Scale temporal GraphRAG for necessity

A knowledge-graph layer mapping payer policies to clinical ontologies, with the **date of service as a hard filter**.

OPTION 2

Deploy prior-guided claim routing using Real World Evidence (RWE)

A Bayesian orchestrator using **OIG-LEIE overrides and facility-network terms** to route fast or deep.

OPTION 3

Scale peer-baseline profiling

Unsupervised agents using **CMS Part B baselines** and specialty z-scores, without labeled fraud data.

These three options are exactly what the proof of concept implements next.

FROM RECOMMENDATION TO RUNNING CODE

The prototype provides PoC for all 3 options

OPTION 1

Temporal GraphRAG necessity**Necessity agent**

retrieves the policy in force on the claim's date of service

OPTION 2

Prior-guided routing**Orchestrator**

Bayesian provider risk + LEIE override decide fast vs deep

OPTION 3

Peer-baseline profiling**Anomaly agent**

specialty z-score against CMS-style peer baselines

Plus, a deterministic coding agent and a reviewer that turns findings into an action; five agents, running end-to-end.

A demo PoC, not a fully functional MVP

● DATA

Runs on **synthetic claims that match real schemas of** CMS Part B and the OIG exclusion list (LEIE).

● SCOPE

Human-in-the-loop decision making. It is not an autonomous payer and does not retrain itself in the current version; the reviewer makes every final call.

● METHOD

Hard parts are **simplified**: a small policy corpus and peer baselines, built on real libraries scikit-learn, networkx, and LangGraph. Graph-RAG full implementation and self-training to be explored in the internship

WHAT TO DESIGN AGAINST

Three threats to manage

01

“Black box” scrutiny

Automated denials **without traceable reasoning** risk provider abrasion, appeals, and penalties.

02

Temporal & semantic hallucination

Naive RAG without date-of-service guardrails can apply **expired policies**, producing invalid denials.

03

Model & agent drift

Agent logic degrades on new edge cases and **evolving fraud**, requiring ongoing validation.

The mitigations are the same properties the design optimizes for: traceability, temporal guardrails, and continuous validation.

From analysis to a defensible decision

The report makes the case for coordinated agents; the prototype proves it turning a risk score into an explained action, tied to the right rule on the right date, with a human in charge of the money decisions.

